From Form to Function and Back Again

Tokenisation, Morphology, and Power-Law Statistics in the Atomisation of Language

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May 2026

Abstract

Tokenisation, the process by which large language models decompose text into atomic units before processing, is widely treated as an engineering preliminary rather than as a substantive theoretical object. We argue against that framing. The choice of atomisation materially shapes the geometric and statistical properties of the resulting representation, and the disagreement between compression-driven tokenisation (Byte-Pair Encoding) and morphologically motivated segmentation is structured rather than incidental. We develop the argument through a graph-theoretic vocabulary of mapping types, an analogy with structural chemistry in the spirit of Coxeter's combinatorial treatment of molecular graphs, and an empirical analysis of three atomisations of a 2-million-token English corpus. Maximum-likelihood power-law fits using the Clauset–Shalizi–Newman procedure show that all three atomisations produce Zipfian tails with measurably different exponents ($\alpha_{\text{words}} \approx 1.64$, $\alpha_{\text{Morfessor}} \approx 2.07$, $\alpha_{\text{BPE}} \approx 2.12$). Tail-restricted Kolmogorov–Smirnov distances reveal a striking convergence between the two compression-driven atomisers ($D = 0.25$) compared with their distance from whole-word frequencies ($D \approx 0.98$). We discuss the methodological care required in applying KS-style tests to heavy-tailed distributions and argue that the underlying form-function inseparability described by the chemistry analogy is the same phenomenon that linguistics describes through Zipf's law: a structural signature of systems whose constituent parts are not independently specifiable.

Keywords: tokenisation; Byte-Pair Encoding; Zipf's law; power-law distributions; Kolmogorov–Smirnov test; morphological segmentation; scale invariance.

1 Introduction

A linguist reading a sentence and a transformer language model processing the same sentence are not engaging the same object. The linguist sees morphemes (the smallest meaning-bearing units), arranged into a phrase-structure tree. The transformer sees tokens, atomic strings produced by an algorithm whose objective is compression rather than meaning. The atomisations are different in principle and, as we shall show, measurably different in practice.

This paper argues that the difference matters. The atomisation determines the statistical geometry of the representation space, the boundaries at which information is lost, and the shape of the power-law distribution that governs both linguistic and computational behaviour at scale. The disagreements between compression-driven and morphologically motivated atomisation are systematic and pedagogically illuminating.

We develop the argument in four stages. Section 2 introduces a graph-theoretic vocabulary for the kinds of mismatch between linguistic and computational atomisations. Section 3 draws an analogy with structural chemistry through the lens of combinatorial graph theory in the tradition of Coxeter [1]. Section 4 argues that the move from classical to quantum chemistry, in which electrons cease to belong to particular bonds and become delocalised over molecular orbitals, mirrors the move from morphological to distributional accounts of meaning that transformers operationalise. Section 5 develops the statistical core: power-law distributions are the natural mathematical language of systems in which form and function are inseparable, and Zipf's law [3] is the linguistic instance of a much wider phenomenon. Section 6 presents an empirical study of three atomisations of the same corpus, with appropriate methodological care for the well-known statistical pitfalls of testing power-law distributions with Kolmogorov–Smirnov-style tests [5, 6].

The interactive supplementary materials which is an HTML dissection of tokenisation [32] and an empirical companion implementing the Zipf analysis [33] accompany this paper.

2 Mapping Types Between Linguistic and Computational Atomisation

Consider two parallel decompositions of an English word. The linguist decomposes *unbelievable* into three morphemes: the negating prefix *un-*, the verbal root *believe*, and the adjectival suffix *-able*. A standard Byte-Pair Encoding tokeniser, by contrast, decomposes the same word into the two substring tokens `unbel` and `ievable`. The two decompositions disagree both on the number of pieces and on where the seams fall. The morphological seam between *believe* and *-able* is invisible to the tokeniser; the tokeniser's seam, between `unbel` and `ievable`, has no morphological significance.

To organise such disagreements we introduce a small graph-theoretic vocabulary. Picture two graphs sitting one above the other. The lower graph has morphemes as nodes and edges connecting morphemes that co-compose into words. The upper graph has tokens as nodes and edges connecting consecutive tokens within words. The interesting question is the structure of the map between them.

Definition 1 (Atomisation map)**.** Let W be a word in some natural language with an established morphological analysis. Let $M(W) = (m_1, m_2, \ldots, m_k)$ be the sequence of morphemes constituting W, and let $T(W) = (t_1, t_2, \ldots, t_\ell)$ be the sequence of tokens produced by some tokeniser when applied to W. The *atomisation map* $\phi_W : M(W) \to \mathcal{P}(T(W))$ assigns to each morpheme the set of tokens whose character-spans overlap that morpheme.

This permits a small classification of disagreement types, borrowing algebraic vocabulary loosely.

Type I (isomorphism). The map ϕ_W is a bijection and each morpheme corresponds to exactly one token sharing its character span. Examples: *walk*, *cat*. This case is statistically rare; it occurs when the morpheme is itself a high-frequency string that earned a token slot during BPE training.

Type II (homomorphism, many-to-one). Multiple morphemes collapse into a single token. Example: *redo* is the morpheme sequence *re-* + *do*, but a typical BPE tokeniser emits the single token `redo`. The morphological seam is invisible to downstream processing; the model must re-learn the compositionality from co-occurrence statistics.

Type III (one-to-many, fragmenting). A single morpheme is shattered across multiple tokens. Example: in *unbelievable* the morpheme *believe* is fragmented across the tokens `unbel` and `ievable`. The morpheme has no surviving representation in the token sequence.

Type IV (context-dependent, non-functional). The map ϕ_W depends on the broader context of W as well as on W itself. The same morpheme is treated differently in different surroundings. Example: *bio-* appears as a single token in *biology* (`biology`), as a clean prefix in *biochemistry* (`bio` | `chemistry`), and is shattered across an unrelated boundary in *biomechanical* (`biom` | `echanical`). This is the deepest mismatch and the case that most strongly motivates the chemistry analogy below.

The four types are not exhaustive but they capture the structurally interesting cases. The empirical analysis in Section 6 will demonstrate that Types II–IV constitute the substantial majority of the off-diagonal mass when BPE and morphological segmentation are compared at corpus scale.

3 Coxeter, Combinatorial Chemistry, and the Alkene Analogy

The vocabulary of mapping types is suggestive but it does not yet explain why the disagreements have a particular character. For that we turn to an analogy from structural chemistry which, properly developed, provides the right intuition for what is happening geometrically.

H. S. M. Coxeter and the broader tradition of combinatorial geometry that he influenced treated molecules as labelled graphs in which nodes carry atomic identities and edges carry bond multiplicities [1, 2]. This graph-theoretic framing of chemistry is more than a notational convenience. It captures the central empirical fact that two molecules sharing the same molecular formula—the same vertex labels with the same multiplicities—can have entirely different chemical and physical properties depending on the placement of edges and the geometry of their arrangement.

The alkenes provide the canonical illustration. The molecular formula C_4H_8 admits at least four distinct molecules: *1-butene*, *cis-2-butene*, *trans-2-butene*, and *isobutylene*. They share atom counts; they differ in double-bond placement and substituent arrangement. The differences in their measurable properties (boiling point, dipole moment, reactivity profile) are not small. *Position determines identity*—an aphorism we shall reach for again.

3.1 Two Greek Roots, Two Distinct Relationships

Before applying the chemistry framing it is worth fixing vocabulary. Two Greek-derived terms appear repeatedly in this analysis and they describe distinct relationships:

- *iso-mer* from *ī́sos* (“equal”) + *méros* (“part”): *same parts, different arrangement.* In chemistry, two compounds are isomers when they share an identical molecular formula but differ in the bond-arrangement of their identical atoms. The four C_4H_8 alkenes are isomers of one another.

- *iso-morph-ic* from *īsos* + *morphē* ("form") + adjectival ending: *same form, possibly different constituents*. In algebra, two structures are isomorphic when there exists a structure-preserving bijection between them, regardless of what their elements individually are.

The chemistry usage is the more demanding constraint. Isomerism requires identical constituents in different arrangements; isomorphism merely requires that the relational structure match under some relabelling. The two should not be conflated.

We will need a third concept too: *substructure recurrence*. Many molecules contain the same chemical group (a methyl, a hydroxyl, a phenyl ring) embedded in different larger structures. The recurring substructure is not an isomer of itself; it is a shared building block whose properties depend on what surrounds it. This will turn out to be the right analogue for the linguistic phenomenon we want to describe.

3.2 Substructure Recurrence: Methyl Groups and the bio- Family

Consider three small molecules: methanol (CH_3OH), toluene (C_6H_5–CH_3), and acetic acid (CH_3–COOH). All three contain a methyl group, the substructure CH_3. The three molecules are not isomers; they have different molecular formulas and different atom counts. They share a recurring substructure appearing in three different chemical environments. The methyl group's electronic and reactive behaviour differs measurably across the three: in methanol it is an alkyl group adjacent to a hydroxyl; in toluene it is an alkyl substituent on an aromatic ring whose hyperconjugation effects donate electron density into the ring; in acetic acid it is the alpha-carbon of a carboxylic acid. The substructure is recognisable in each case but its role, and hence its contribution to the molecule's behaviour, depends on its neighbours.

The same description applies to the morpheme *bio-* across linguistic compounds. Consider three English words sharing the prefix:

$$\begin{aligned} \textit{biology} &\longmapsto \texttt{biology} \\ \textit{biochemistry} &\longmapsto \texttt{bio}\,|\,\texttt{chemistry} \\ \textit{biomechanical} &\longmapsto \texttt{biom}\,|\,\texttt{echanical} \end{aligned}$$

These three words are *not* isomers in the strict chemistry sense. They have different total letter counts and different constituent material beyond the shared prefix. What they share is the substructure *bio-*. The interesting observation is that this shared substructure receives three different tokeniser-level treatments depending on what follows it: absorbed into a single token in the first case, preserved as a clean prefix in the second, shattered across an unrelated boundary in the third.

This is not isomerism. It is the analogue of methyl-in-toluene versus methyl-in-acetic-acid: the same subgraph appearing in three different molecular environments, with three different effective behaviours. The morpheme *bio-* has no stable representation in the tokeniser's vocabulary; what survives of *bio-* is determined entirely by its neighbours, exactly as the methyl group's electronic environment is determined by its neighbours. This is the correct chemistry analogue for Type IV in Section 2.

3.3 Where Genuine Isomerism Does Apply

The bio-words illustrate substructure recurrence rather than isomerism. There is, however, a separate linguistic phenomenon for which the strict isomerism analogy holds, and it is worth naming it to keep the two distinct.

Consider letter-level anagrams: *unite* and *untie* share the same five letters $\{u, n, i, t, e\}$ in different orders. They are isomers of each other in the strict chemistry sense: identical molecular formula (the multiset of letters), different graph (the sequential order). The same applies to *traps / parts / strap / sprat*, or *stressed / desserts*.

Empirically these letter-isomers do not provoke distinguishable tokeniser behaviour for an interesting reason: most short common anagrams are themselves frequent enough to earn single tokens during BPE training, so they collapse onto a token-level representation that is one piece long regardless of the letter order. Where letter-isomerism does become observable at the tokeniser level is in low-frequency anagrams: *stressed* and *desserts* share letters but the latter (a plural common noun) is more frequent than the former (a verb's past form), and their BPE segmentations differ. Investigating this systematically is beyond the scope of the present paper but represents a clean follow-up question.

A separate phenomenon worth distinguishing is *homography*: two words spelled identically but pronounced or interpreted differently in context, such as *lead* (the metal) versus *lead* (the verb). The graph of letters is identical in both cases; the meaning is disambiguated by context only. This is neither isomerism (no rearrangement) nor substructure recurrence (the words coincide entirely). The closest chemistry analogue is the use of *resonance structures* to describe a single molecule whose electron distribution is best represented as a superposition of distinct Lewis-structure depictions. Benzene's two Kekulé structures are the textbook case: identical atoms, identical bonds, different conventional drawings of an underlying delocalised reality. We mention this analogy only to avoid confusion. The central chemistry analogue of this paper is substructure recurrence rather than isomerism or resonance.

3.4 The Word Isomorphic Itself, Tokenised

A small but pedagogically pleasing observation closes this section. The word *isomorphic* is itself a textbook case of computational inequivalence between linguistic and tokeniser-level atomisation. A linguist sees a clean three-piece compound: *iso-* + *morph-* + *-ic*. A typical BPE tokeniser produces approximately `iso | m | orphic`, slicing straight through the morpheme *morph*. The word that names structural equivalence is itself a casualty of structural inequivalence at the tokeniser level. The lesson recurses.

4 Form, Function, and the Classical-to-Quantum Transition

The chemistry analogy can be pushed further. The history of chemistry from the late nineteenth century to the present can be read as a transition from a *classical* picture, in which atoms have fixed valences and bonds are localised between specific pairs of atoms, to a *quantum* picture, in which electrons are delocalised over molecular orbitals and bonds are statistical features of an electron-density distribution rather than discrete geometric entities.

We argue that an analogous transition is currently underway in linguistics. The classical picture (generative grammar in the Chomskyan tradition) treats language as built from discrete morphological and syntactic atoms combined according to fixed rules. The modern picture (transformer-based language models with their distributional semantics) treats word meaning as a position in a high-dimensional vector space whose geometry is shaped by co-occurrence statistics. In the modern picture there is no fact-of-the-matter about what a word "means" independent of its context; meaning is a learned function of the surrounding distribution.

Table 1 summarises the parallel.

Era	**Chemistry**	**Linguistic counterpart**
Classical / structural	Lewis structures, fixed bonds, localised valence; molecular formula plus structural diagram suffices.	Generative grammar; discrete morphological categories; hard combinatorial rules; fixed lexical entries.
Modern / quantum	Molecular orbitals; delocalised electron density; resonance; attention as long-range coupling between distant sites.	Transformer semantics; distributed vector representations; context-dependent meaning; attention as long-range dependency.

Table 1. The two-tier picture. The classical picture localises properties; the modern picture spreads them.

The parallel is not metaphorical. In both cases the modern picture preserves the data of the classical picture as a special case (one can still draw Lewis structures of benzene; one can still parse sentences) but reveals that the classical entities are limit-cases of a more distributed underlying reality. Bonds in benzene look localised when you draw them but the electrons are spread evenly around the ring. Words in a sentence look discrete when you transcribe them but their effective representations in a transformer are spread over a network of attention weights.

4.1 A Caveat on the Statistical Mechanism

A reader familiar with quantum mechanics might expect the analogy to extend to specific exclusion principles such as Pauli statistics. We resist that extension. Pauli's exclusion principle is a symmetry constraint on fermionic wavefunctions arising from particle indistinguishability and half-integer spin; it produces Fermi–Dirac statistics and is responsible for the structure of the periodic table. The statistical phenomenon we describe in Section 5, by contrast, is a power law, and power laws arise from a quite different family of mechanisms: scale invariance under coarse-graining, preferential attachment, multiplicative noise under multiplicative growth, and optimisation under multiplicative constraint [6, 7].

The right physical analogue for Zipf-like behaviour in language is not Pauli but the renormalisation group [8], the framework developed in statistical physics for understanding why systems near critical points exhibit scale-invariant behaviour and why power laws are the unique distributions invariant under rescaling. The renormalisation group framing connects the form–function inseparability of Section 3 to the empirical Zipfian regularity of Section 5 through a single substantive principle: in systems whose elements are functionally interdependent across all scales, the only distributions that are stable under rescaling are power laws. We develop this connection in the next section.

5 Zipf's Law as the Signature of Form-Function Inseparability

5.1 The Empirical Regularity

Zipf's law states that, when units of natural language are ranked by their frequency in some sufficiently large corpus, the frequency f_n of the nth-ranked unit is approximately inversely proportional to its rank:

$$f_n \propto \frac{1}{n^s}, \qquad s \approx 1. \tag{1}$$

The relationship has been observed across hundreds of languages, across corpora spanning many centuries of written material [4], across spoken and written registers, and across many non-linguistic domains: city populations [9, 10], firm sizes [11], citation counts [12], and biological transcriptomes [13]. The robustness of the regularity has invited many explanations: the principle of least effort [14], preferential attachment [15, 16], monkey-typing artefacts [17, 18], and entropy-maximisation under constraint [19]. None has achieved consensus.

5.2 The Connection to Form-Function Inseparability

The renormalisation group framing suggests an interpretation that unifies the chemistry analogy of Section 4 with the empirical Zipfian regularity. Power laws are the unique scaling solutions of the rescaling equation $f(\lambda x) = \lambda^{-s} f(x)$ and arise generically in systems whose statistical structure is invariant under coarse-graining. Such systems share a characteristic feature: their elements are not functionally independent at any scale, and any attempt to specify the role of an individual element requires reference to its position in the wider system. This is precisely the form-function inseparability captured by the chemistry analogy. The same principle operates in metabolism, where Kleiber's law $B \propto M^{3/4}$ links basal metabolic rate B to organism mass M across more than twenty orders of magnitude in mass [20, 21] and where the units (cells, organs) are not specifiable independent of the transport network that supplies them.

We do not claim that linguistic Zipf and biological Kleiber share a generative mechanism in any narrower sense; that claim has been explicitly contested by Newman and others [6] and remains open. We claim only that they share a structural characteristic—heavy-tailed distributions over heterogeneous populations whose elements are functionally interdependent—and that this characteristic is what makes a power-law description appropriate in both cases.

5.3 Why Tokenisation Depends on Zipf's Law

The argument develops in a specific direction relevant to BPE tokenisation. Byte-Pair Encoding is a compression algorithm whose effectiveness presupposes Zipf-like statistics. The merge rule preserves symbol pairs whose co-occurrence is unusually frequent relative to independence. If language followed a uniform distribution rather than a power law, the merges would not concentrate around reusable substrings and the resulting token vocabulary would degenerate into noise. BPE succeeds because language already contains deep statistical asymmetries, and the asymmetries are themselves the signature of the form-function inseparability described above. In this sense the tokeniser is not merely a preprocessing engineering choice; it is parasitic on a deep regularity of the linguistic substrate.

6 Empirical Study: Three Atomisations of the Same Corpus

6.1 Methods

We constructed a corpus of $N = 2{,}000{,}000$ English word tokens by sampling from the `wordfreq` aggregated frequency dictionary [22], which combines Wikipedia, Project Gutenberg, contemporary news, subtitle corpora, Reddit, and Twitter. The same running text was then atomised three ways.

Two terms worth unpacking before we proceed

The three atomisations in this section make repeated use of two words that are second nature in computational linguistics and opaque to almost everybody else. Since the spirit of this paper is to take vocabulary seriously, both deserve a small unpacking.

Whitespace. The term names a category of characters in text that occupy visual space without producing visible marks: the ordinary space between words, the tab, the line break, the carriage return, occasional more exotic Unicode separators. The term comes from typesetting and pre-digital printing, where compositors literally placed thin metal strips between letterforms to produce gaps; the strips left empty space on the white printed page. Early computer scientists adopted the word because the same characters play the same role in source code: they are the seams between meaningful tokens. Today *whitespace* in programming and text-processing means any run of one or more space-equivalent characters, treated as a delimiter rather than as content.

A *whitespace tokeniser* is therefore the simplest possible text segmenter: it scans the text character by character and emits a token boundary every time it sees one or more whitespace characters. The string `the␣cat␣sat` becomes the three tokens `the`, `cat`, `sat`. This procedure has the great virtue of being trivial to implement and the corresponding vice of being linguistically uninformative: it locates the gaps that the writer or printer chose to put in, nothing more.

Morphology and morpheme. The term *morphology* in linguistics is borrowed directly from the same Greek root that we met in §3: *morphē* ("form") + *logos* ("study, account"), *the study of form*. The word entered modern science through nineteenth-century biology (Goethe coined *Morphologie* in 1796 for the study of the forms of organisms; D'Arcy Thompson's *On Growth and Form* from 1917 is the most famous English exposition) and was adopted by linguistics in the late nineteenth century to name the study of the internal form of words.

A *morpheme* is the minimal unit of linguistic form that carries meaning. The word *walking* contains two morphemes: *walk* (the verb root) and *-ing* (the marker of the present participle). The word *unhappiness* contains three: *un-* (negation), *happy*, and *-ness* (abstract noun formation). Morphemes are the linguistic analogue of atoms in chemistry: the smallest pieces with stable identity out of which words are built. They are not the same as syllables (*walking* has two syllables but two morphemes that happen to align; *unhappiness* has four syllables and three morphemes that do not align with them).

A *morphological segmentation* of a word is its decomposition into the constituent morphemes. The morphological segmentation of *walking* is ⟨*walk*, *-ing*⟩; the morphological segmentation of *unhappiness* is ⟨*un-*, *happy*, *-ness*⟩. A *morphologically motivated tokeniser* is one whose token boundaries coincide, as closely as it can manage, with morpheme boundaries.

The contrast that drives the rest of the paper. Whitespace tokenisation finds the seams the writer left between words. Morphological tokenisation finds the seams a linguist would identify between meaning-bearing units inside words. They operate at completely different scales. Whitespace ignores everything inside a word. Morphology ignores the spaces between words and works exclusively inside them. Subword tokenisation as practised by Byte-Pair Encoding sits awkwardly between the two: it works on the same character-level material as morphology, but its segmentation criterion is statistical compression rather than linguistic meaning, and its boundaries therefore coincide with morphological boundaries only by accident. The remainder of this section examines the three approaches in turn and what each produces on the same corpus.

1. Whitespace words

The classical Zipf baseline. The corpus is split on whitespace boundaries (as defined immediately above) to recover individual word-form types. This is what Zipf himself examined in 1949 and what most of the historical literature on Zipf's law studies. It is worth being explicit about what this baseline does and does not represent.

A whitespace tokeniser treats each distinct *surface form* as a unit. The forms *walk*, *walks*, *walked*, *walking* are four separate types because they are four distinct strings between whitespace boundaries, even though a linguist would say they are four inflected forms of one underlying verb (the lemma *walk*). The forms *cat* and *Cat* are also separate by the same logic; we cast to lower case here to absorb that artefact, which substantially reduces noise but does not eliminate it. Punctuation, possessives, and contractions become boundary problems: *don't* can become one token or two depending on the splitter; *John's* similarly. The `wordfreq` corpus we work with has already been normalised in standard ways before exposure to us, so most of these issues are resolved by the upstream pipeline.

Crucially, the whitespace baseline is *neither morphological nor sub-word*. It does not recognise that *walking* contains the morphemes *walk* and *-ing*, and it does not split words into smaller pieces of any kind. A linguist would call it *word-form-based* rather than morphologically analysed. The vocabulary is therefore as large as the inflectional and derivational richness of the language allows; for English on this corpus we observe 68,555 distinct word types after filtering. For a morphologically richer language such as Finnish or Turkish the same baseline would explode into hundreds of thousands or millions of types, which is one of the practical reasons modern language models do not use this atomisation.

2. BPE tokens

A Byte-Pair Encoding tokeniser was trained *on this corpus* (vocabulary cap 16,000) using HuggingFace `tokenizers` [25], then applied back to the corpus. BPE [23, 24] begins from a character-level vocabulary. It then iteratively scans the corpus for the most frequent adjacent pair of symbols and merges that pair into a new vocabulary entry, repeating the process until the vocabulary cap is reached. The result is a vocabulary of variable-length subword pieces in which frequent character sequences (such as `the`, `ing`, `tion`) are absorbed into single tokens while rare character sequences are left fragmented. There is no linguistic input to the procedure; the only signal is co-occurrence frequency.

Training BPE locally rather than using a third-party-trained encoding such as OpenAI's `cl100k_base` is methodologically important here. A pretrained BPE was trained on text we cannot inspect, so any difference between `cl100k_base` behaviour on our corpus and the corpus's own word frequencies could come from BPE's algorithm or from train-test mismatch between the BPE's training set and ours. By training BPE on the same corpus we then apply it to, we eliminate the train-test mismatch and any difference observed must come from the BPE algorithm itself.

3. Morfessor morphs

An unsupervised morpheme-like segmentation was learned by Morfessor 2.0 Baseline [26, 27] from the corpus's word-frequency dictionary. Morfessor deserves a slightly fuller explanation because its relationship to true linguistic morphology is the subtler of the three.

What Morfessor does. The algorithm takes as input a list of word types and their frequencies (here, our 68,555 filtered word types from the materialised corpus). It then searches for a segmentation of each word into a sequence of substrings, called *morphs* in the Morfessor literature, subject to two competing pressures. The first pressure favours short codes for the collection of morphs (a small inventory). The second favours short codes for the segmentations themselves (each word represented by few morphs). The objective is a *Minimum Description Length* (MDL) loss combining the cost of the morph lexicon with the cost of describing the corpus given that lexicon. Optimising this loss implicitly produces a segmentation that reuses common subword chunks across many words, much as a linguist's morphological analysis would reuse *-ing* across many verbs.

How it relates to morphemes. Morfessor's *morphs* are designed to approximate *morphemes* but are not guaranteed to coincide with them. The MDL objective is purely information-theoretic and contains no explicit linguistic prior, no grammar, no part-of-speech information, and no allomorphy handling. It therefore makes characteristic errors. *injustice* is correctly segmented as *in* + *justice*. *walking* is correctly segmented as *walk* + *-ing*. But *anthem* (a single Greek-derived morpheme) is incorrectly segmented as *an* + *them* because *an* and *them* both occur frequently as units across the vocabulary and the MDL objective rewards reuse over linguistic faithfulness. *justin* (a proper noun) is incorrectly segmented as *just* + *in* for the same reason. Morfessor is therefore best understood as an *unsupervised algorithmic proxy* for morphological segmentation: it agrees with linguistic morphology on the easy cases (transparent compounds, productive affixation) and disagrees on the hard cases (etymologically opaque vocabulary, proper nouns, borrowings).

Why we use it anyway. A truly morphologically annotated corpus such as CELEX [30] would be the gold standard for this comparison, and we recommend that follow-up extension in the discussion. Morfessor's advantage for the present analysis is that it runs unsupervised on the same input that BPE sees, which makes the comparison *algorithm-versus-algorithm* rather than *algorithm-versus-curated-resource*. The comparison we are interested in is whether two unsupervised atomisers with different objective functions (compression in the case of BPE, MDL in the case of Morfessor) produce statistically distinguishable atomisations of the same text. They do, and the form their disagreement takes (BPE shatters rare words; Morfessor splits frequent compounds) is the substantive empirical finding of the paper. Replacing Morfessor with CELEX-grade segmentations would shift the specific numbers but would not, we argue, alter the qualitative pattern of disagreement.

Power-law fitting

For each atomisation we compute the rank-frequency distribution and fit a discrete power law to the tail using the Clauset–Shalizi–Newman [5] maximum-likelihood procedure as implemented in the `powerlaw` Python package [28]. The procedure also identifies the lower cutoff $x_{\min}$ above which the power-law fit is valid; below that cutoff the data may behave differently.

6.2 Power-Law Exponents

Atomisation	α	σ	$x_{\min}$	n_{tail}	D_{fit}
Whitespace words	1.643	0.002	1	68,555	0.015
Morfessor morphs	2.069	0.028	161	1,490	0.017
BPE tokens	2.122	0.031	212	1,328	0.018

Table 2. Maximum-likelihood power-law fits using the Clauset–Shalizi–Newman procedure. σ is the bootstrap standard error. D_{fit} is the Kolmogorov–Smirnov distance between the empirical distribution and the fitted power law within the tail.

The three atomisations produce *measurably different* power-law exponents (Table 2). The two algorithmic atomisers (BPE, Morfessor) yield steeper tails ($\alpha \approx 2.07$–2.12) than the underlying word-frequency distribution ($\alpha \approx 1.64$). The differences in α exceed their standard errors by more than an order of magnitude.

The interpretation is direct: a compression-driven atomiser, by construction, gives short codes to frequent items and longer codes to rare items. The resulting frequency distribution must therefore concentrate probability mass more sharply on the common items than the underlying word-frequency distribution does. *Compression produces steeper Zipfian tails as a structural consequence of its objective.* BPE and Morfessor both exhibit this property because both have compression-flavoured objectives, despite their algorithmic differences.

The KS distances between each empirical distribution and its fitted power law (final column, Table 2) are uniformly small ($D \approx 0.015$–0.018), indicating that the power-law form is itself a reasonable description of the tail in all three cases. The differences between the atomisations are differences in α, not differences in whether they obey power laws at all.

6.3 Methodological Note on KS Tests for Heavy-Tailed Data

The standard two-sample Kolmogorov–Smirnov test, while non-parametric and distribution-free, has a known weakness for heavy-tailed data: the test statistic

$$D = \sup_x |F_1(x) - F_2(x)| \tag{2}$$

is dominated by the bulk of each distribution where the empirical CDFs are rapidly increasing, and is comparatively insensitive to differences in the heavy tail where the interesting power-law structure lives. Two genuinely different power-law distributions with similar medians can register a small KS distance; two otherwise-similar power laws with different medians can register a large one. Clauset, Shalizi and Newman [5] discuss this issue at length and recommend tail-restricted variants and Anderson–Darling-style alternatives for power-law work.

We address the issue with a triple analysis: full-distribution two-sample KS, tail-restricted KS computed only on values above each pair's $x_{\min}$, and the k-sample Anderson–Darling statistic which weights the tail more heavily by construction. Anderson–Darling is computed via `scipy.stats.anderson_ksamp`.

6.4 Pairwise Distribution Distances

Pair	$\mathbf{KS_{full}}$	$\mathbf{KS_{tail}}$	**AD** A^2
Words vs. BPE	0.757	0.985	21,054
Words vs. Morfessor	0.369	0.980	12,460
BPE vs. Morfessor	0.562	**0.248**	7,698

Table 3. Pairwise distribution comparisons. The reported numbers are the test statistics (D for KS, A^2 for Anderson–Darling), which quantify the magnitude of the distributional difference. We do not report p-values: at the sample sizes involved here (tens of thousands of distinct types) any non-trivial difference is detected with overwhelming statistical significance, so a reported p would only confirm that the test had the power to detect what is already visible in the test statistic. The boldface entry highlights the central empirical finding: under tail-restricted KS, BPE and Morfessor are roughly four times closer to each other than either is to whole-word frequency.

A note on p-values at this sample size. The two-sample KS statistic D is converted to a p-value via an asymptotic formula whose dominant term is $\exp(-2D^2 n_1 n_2/(n_1+n_2))$. With n_1, n_2 in the tens of thousands and D above 0.2, the exponent runs to several hundred negative or worse, and the resulting p-value underflows to zero in standard double-precision arithmetic. Reporting "$p < 10^{-300}$" in this regime is not a finding; it is a numerical artefact of the test being grossly overpowered for the question. The substantive content lies in the magnitude of D and A^2, which quantify how far apart the distributions are, rather than in p-values that are guaranteed to be vanishingly small whenever the distributions differ at all.

Three observations follow from Table 3, all of which we take as substantive.

Full-distribution KS reflects vocabulary mismatch. Words versus BPE produces the largest KS distance ($D = 0.757$) because the underlying support sets are of fundamentally different sizes: 68,555 word types versus 15,765 used BPE types under a vocabulary cap of 16,000. KS captures this strongly but the captured difference is partly an artefact of the cap and not entirely a difference in distributional shape.

Words versus Morfessor is the closest full-distribution pair. $D = 0.369$. Morfessor's segmentation, being closer to morphologically motivated, leaves the rank-frequency shape closer to whitespace words than BPE's aggressive merging does.

The tail-restricted comparison inverts the picture. Restricting attention to values above each pair's $x_{\min}$, words versus BPE and words versus Morfessor both jump to $D \approx 0.98$ because the tail samples are at very different scales. But BPE versus Morfessor *collapses* to $D = 0.248$, roughly four times smaller than the corresponding full-distribution distance. *The two compression-driven atomisers, restricted to their power-law tails, produce statistically similar distributions even though their individual segmentations differ.* This is the formal evidence for the structural claim made in Section 5: compression-driven atomisation has a characteristic Zipfian signature distinct from the morphologically grounded one.

The Anderson–Darling statistic broadly confirms the KS picture with the expected stronger numerical separation. As with KS, the relative rather than absolute magnitudes are the interpretively useful information.

6.5 Heaps' Law and Vocabulary Saturation

The companion observation to Zipf's law is Heaps' law: vocabulary size V grows with corpus size N as

$$V \approx KN^{\beta}, \tag{3}$$

with $\beta \in (0, 1)$ for natural-language text [29]. We fit Heaps' law by linear regression on log-log axes for each atomisation (Table 4).

Atomisation	β	K	R^2
Whitespace words	0.681	5.07	0.9898
Morfessor morphs	0.623	7.79	0.9720
BPE tokens	0.538	14.84	0.9265

Table 4. Heaps' law fits. Lower β indicates more aggressive vocabulary saturation as the corpus grows.

The Heaps exponent β provides a direct measure of vocabulary openness. BPE's low value ($\beta = 0.538$) reflects its fixed vocabulary cap: once the cap is saturated, no new types can appear regardless of corpus size. Whole-word vocabulary continues to grow ($\beta = 0.681$) because the corpus always exposes new rare items. Morfessor's intermediate value ($\beta = 0.623$) reflects its unconstrained but reuse-favouring objective: it has no cap, but its MDL prior reuses morph types aggressively as the corpus grows. The pattern is the Zipfian story told in a different statistic: the algorithmic atomisers concentrate information more aggressively than naive whole-word counting does.

6.6 Where the Atomisers Disagree

The pairwise distances in Table 3 are aggregate statistics. To complement them we examine the joint distribution of pieces-per-word for individual word types. For every word in the corpus we record both its BPE piece count and its Morfessor piece count. Most probability mass concentrates on $(1, 1)$ (both atomisers keep the word whole). The off-diagonal mass falls into two characteristic patterns.

BPE shatters where Morfessor preserves. Rare proper nouns and Latinate vocabulary that Morfessor recognises as monomorphemic are aggressively fragmented by BPE:

- *schizophrenia* $\rightarrow$ `sch` | `iz` | `oph` | `ren` | `ia` ($Morfessor$: `schizophrenia`)
- *Czechoslovakia* $\rightarrow$ `cze` | `cho` | `slovak` | `ia`
- *Antarctica* $\rightarrow$ `ant` | `ar` | `ct` | `ica`

None of these are linguistically morphemic splits. BPE's algorithm, blind to monomorphemic status, dismantles them into the closest available subword chunks, and the model has to relearn the unity downstream from co-occurrence.

Morfessor splits where BPE preserves. Frequent morphologically transparent compounds are kept whole by BPE (it earned vocabulary slots for them through merging) but split by Morfessor:

- *injustice* → *in* + *justice* (Morfessor); single token (BPE)
- *incoming* → *in* + *coming* (Morfessor)
- *inappropriate* → *in* + *appropriate* (Morfessor)
- *thereby* → *there* + *by* (Morfessor)

These are morphologically clean splits. The compounds are frequent enough that BPE earns a single token for them, but linguistically they are decomposable.

The disagreement is therefore systematic and bidirectional. BPE makes the rare-word shattering error; Morfessor makes the frequent-compound splitting error. Both are wrong from a strict linguistic perspective and both are right from the perspective of their own objective function. This is the empirical realisation, at corpus scale, of Type IV from Section 2.

7 Discussion

We have argued, both by analogy and by direct empirical comparison, that the disagreements between morphologically motivated and compression-driven atomisations of language are structured rather than incidental. Three threads converge.

First, the graph-theoretic vocabulary of mapping types (Section 2) provides a small classification scheme into which observed disagreements fall systematically. The empirical work in Section 6 confirmed that Type IV disagreements (context-dependent fragmentation) constitute the substantively interesting cases at corpus scale, and that they have characteristic asymmetric directions (BPE shatters rare items; Morfessor splits frequent compounds).

Second, the chemistry analogy (Sections 3–4) provides physical intuition for why the disagreements take the shape they do. Position determines identity; the same atoms in different positions are different objects. The analogy with classical-to-quantum chemistry suggests further that the modern distributional picture preserves the classical morphological picture as a limit case while revealing it as an abstraction over a more delocalised substrate.

Third, the empirical Zipf analysis (Section 6) quantifies the structural difference. All three atomisations produce power-law tails, but with measurably different exponents. Tail-restricted KS distances reveal that the two compression-driven atomisers converge on a common statistical signature. Heaps' law exponents make vocabulary saturation directly visible. The methodological discussion of KS for heavy-tailed data (Section 6.3) is a reminder that statistical tests have characteristic blindnesses, and that responsible analysis of power-law data requires explicit treatment of the tail-versus-bulk distinction.

7.1 Open Questions

Several substantive questions remain open. The most important is the underlying mechanistic question: why do linguistic units obey Zipf's law in the first place? The candidate explanations [4, 6, 7] are not yet decisive. Our argument is compatible with any of them. We have only claimed that the form-function inseparability of linguistic units, whatever its ultimate origin, has a power-law signature that BPE-style tokenisation depends on for its operation.

Two further empirical extensions would strengthen the argument. First, replication on a non-English corpus would test whether the specific exponents reported here are language-specific or universal. Second, comparison with a gold-standard morphologically segmented corpus (CELEX [30] or MorphoChallenge [31] data) would test whether the asymmetric disagreement patterns of Section 6 survive the replacement of Morfessor's unsupervised proxy with linguistically authoritative segmentations. We expect both extensions to broadly confirm the qualitative picture but to shift the specific exponents in ways that would themselves be informative.

Acknowledgements

The interactive companion materials [32, 33] were developed in parallel with this paper and substantially influenced its argument. The Anthropic Claude language model was used as a research and writing collaborator throughout, particularly in clarifying the form-function argument in Section 4 and in implementing the empirical pipeline of Section 6.

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